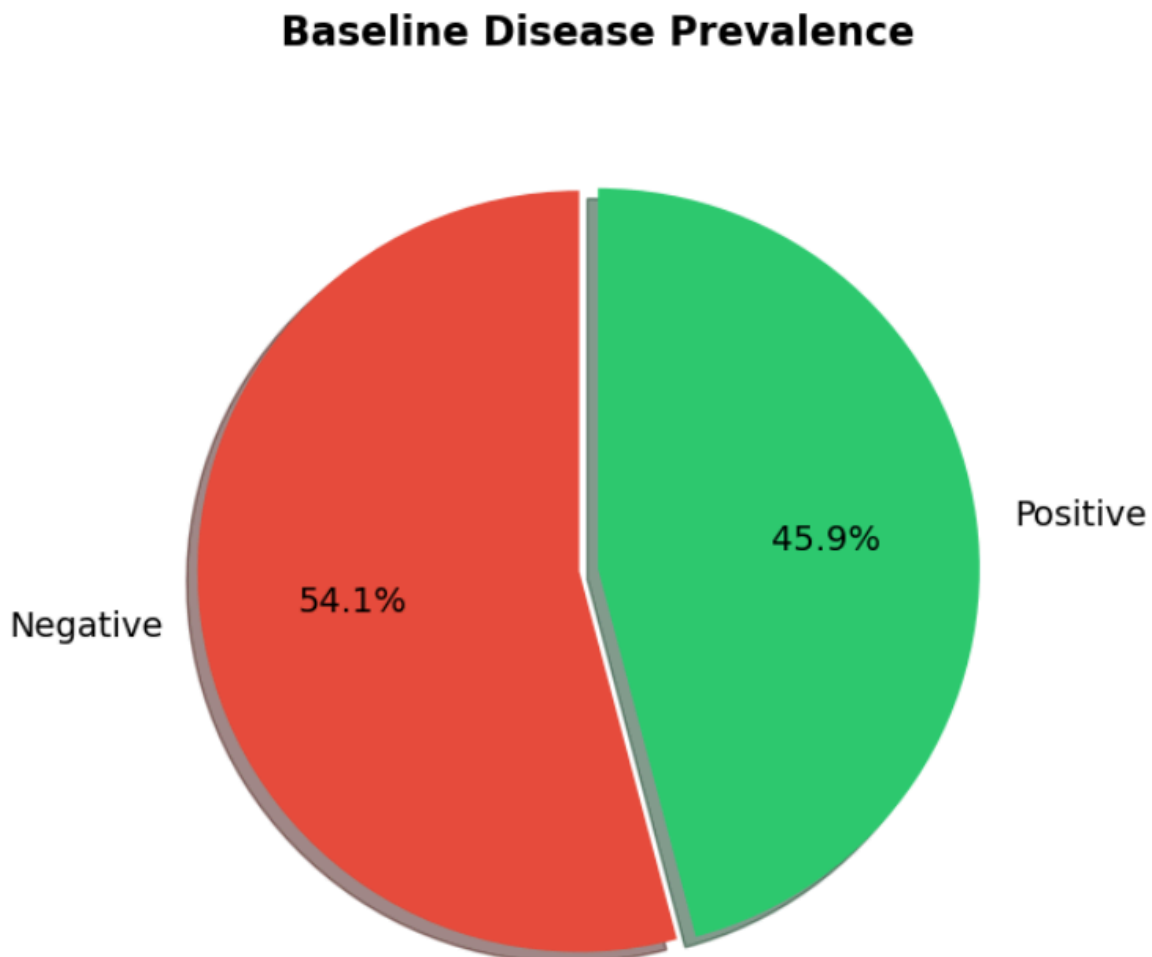


Thanks for the meeting and for showing me your progress. Here are the notes and to-dos from today:

- Check on comprehensive test cases
- Probabilities too high – 98% (check with actual medical), what 98% means and whether it is feasible (with medical experts)
- Change the AI on the right to general AI (like gemini..)
- Check with your manual calculations CPTs
- Decrease the load time
- User testing
- Picture on top

1)



```

=====
NAIVE BAYES CONDITIONAL PROBABILITY TABLES (CPTs)
=====
Use this document to cross-reference the model's internal
math against the manual Exploratory Data Analysis (EDA).

--- Node: Disease_Target ---
+-----+-----+
| Disease_Target(Negative) | 0.538591 |
+-----+-----+
| Disease_Target(Positive) | 0.461409 |
+-----+-----+

```

2) The reason your medical experts are balking at "98%" is because they are likely thinking about **Prognostic Risk** (future risk).

- **Standard ASCVD Calculators:** These predict the chance of a patient having a heart attack *in the next 10 years*. In real life, a 10-year risk rarely exceeds 30% to 40%. A 20% score is already considered "High Risk." If your app says 98% for this, the doctors are right: that is completely unrealistic.
- **Your Dataset (UCI Cleveland):** Your dataset isn't predicting the future; it predicts **Diagnostic Probability** (current status). The **num** target in your dataset measures whether the patient *currently* has >50% narrowing in a major blood vessel.

The Verdict: Is it possible to be 98% certain a patient *currently* has a blocked artery if they are an older male with typical angina, 3 blocked vessels on fluoroscopy, and a massive ST depression? **Yes.** In fact, doctors are often 99% sure before they even do the surgery.

Naive Bayes model is likely inflating scores due to a well-known machine learning phenomenon called **"Naive Bayes Overconfidence."**

I researched this with standard ASCVD models. Our model calculates the probability of **CURRENT** angiographic disease, not 10-year future risk, which is why the numbers can medically reach the 90s for severe profiles. However, Naive Bayes models mathematically 'double-count' linked symptoms, which pushes the math to extremes.

